# Manish R

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## Professional Summary

Applied AI Engineer with 3+ years of experience building production LLM systems end to end, from multi-agent orchestration and RAG pipelines to evaluation, observability, and deployment. Specialized in LangGraph and LangChain agent architectures, MCP tooling, knowledge graphs, and vector search, delivered on Python/FastAPI backends and React/Next.js frontends. Shipped AI and security-critical software for clients across investment research, hospitality, fintech, Bitcoin, and cloud infrastructure, with outcomes including **25% faster** room turnover, **90% less** deployment downtime, and **sub-200ms** semantic retrieval. Strong testing discipline with Pytest, Vitest, Playwright, and blocking LLM eval gates.

## Technical Skills

**Generative AI & Agents**: LangGraph, LangChain, Multi-Agent Systems, Agentic Workflows, RAG, Prompt Engineering, LLM Evaluation, MCP (FastMCP), LiteLLM, LangSmith, LangMem SDK
**LLM Providers**: OpenAI, Anthropic Claude, Google Gemini, Groq, OpenRouter
**Data & Retrieval**: Vector Databases (Qdrant, ChromaDB, FAISS), Knowledge Graphs (Memgraph), PostgreSQL, Redis, MongoDB, SQLite, Supabase, SQLAlchemy, Alembic, ELK Stack
**Languages**: Python, TypeScript, JavaScript, Go, Rust, Solidity, SQL, Bash, Embedded C
**Backend**: FastAPI, Node.js, Express.js, Bun, RESTful APIs, Server-Sent Events (SSE), WebSocket, Microservices, Async Python
**Frontend**: React, Next.js, Tailwind CSS, Zustand, TanStack Query, Browser Extensions (Manifest V3, WXT), Vite, Streamlit
**Cloud & DevOps**: Docker, Kubernetes, GCP, GKE, Terraform, GitHub Actions, GitLab CI, Cloudflare Workers, Vercel, Railway, Prometheus, Grafana, Snyk, Trivy
**Testing & Observability**: Pytest, Vitest, Playwright, End-to-End Testing, LLM Eval Gates, LangSmith Tracing, Accessibility Testing (axe-core)
**Blockchain & Security**: Bitcoin (Taproot, Schnorr, BIP-32/39, bitcoinjs-lib), Self-Custody Wallets, Applied Cryptography (AES-GCM, PBKDF2), Ethers.js, Web3.js, Smart Contracts (Solana Anchor, Solidity), x402 Protocol, ERC-8004, ZK Proofs
**IoT & Embedded**: ESP32, Raspberry Pi, IoT Sensors, Real-time Monitoring
**Practices**: System Design, Client-facing Solution Design, Agile, Technical Leadership

## Experience

### Applied AI Engineer

*Zo House, Whitefield*

- Designed and deployed an agentic workflow system for a leading multi-national hospitality brand managing hundreds of properties (hostels, luxury homes, and boutique hotels) across diverse geographies.
- Architected LangGraph-based orchestration layer for housekeeping and House Captain task routing, priority escalation, and real-time logistics management with zero manual scheduling errors during pilot.
- Achieved a **25% increase** in room turnover speed and a **30% boost** in housekeeping productivity through AI-driven task assignment via LangChain agent pipelines.
- Built FastAPI backend with PostgreSQL and Supabase for scalable multi-property data management; delivered React.js dashboard for operational visibility.
- Tech Stack: LangGraph, LangChain, FastAPI, React.js, PostgreSQL, Supabase, Docker

### Applied AI Engineer

*Atlantis*

- Built and maintained AI-augmented multi-chain bounty monitoring and decentralized funding platforms; resolved critical bugs and shipped features improving system stability and load times by **20%**.
- Integrated LangChain RAG pipelines for on-chain data querying and automated reporting across blockchain networks using Ethers.js.
- Designed and prototyped pay-as-you-go monetization layers using the x402 protocol and ERC-8004 standard for fintech data access use cases.
- Tech Stack: Python, LangChain, React.js, TypeScript, MongoDB, Ethers.js, Express.js, Node.js, IoT Sensors, WebSocket

## Projects

**Multi-Agent AI Due Diligence Platform** **Client Project (NDA)**

- *Tech Stack: Python, FastAPI, LangGraph, LangSmith, LiteLLM, PostgreSQL, Memgraph, Qdrant, Redis, Next.js, React, TypeScript, Docker, GitHub Actions*
- Primary engineer on a multi-agent LLM platform that automates company due diligence, turning a single company name into an auditable, source-grounded recommendation.
- Architected a LangGraph state-machine pipeline (discovery, validation, analysis, synthesis) with specialist scoring agents and a multi-persona council that debates and reconciles findings.
- Designed a FastAPI microservice backend with real-time SSE progress streaming, a knowledge graph, vector retrieval, Redis caching and rate limiting, and per-service fault isolation so one failing datastore never takes down the platform.
- Built a governance layer with data lineage on every result, claim grounding, and an explicit unknown-value contract so missing data is never reported as a score.

- Enforced quality with blocking LLM eval gates for factual correctness, LangSmith tracing, and **290+ test files** across Pytest, Vitest, and Playwright E2E with automated accessibility checks.

**Bitcoin Self-Custody Vault Wallet** **Client Project (NDA)**

- *Tech Stack: TypeScript, React, WXT (Manifest V3), Vite, bitcoinjs-lib, Web Crypto API, Node.js, Express.js, Cloudflare Workers, Vitest, Playwright*
- Sole engineer of a cross-browser (Chrome and Firefox) Manifest V3 wallet extension for Bitcoin self-custody vaults with Taproot Schnorr signing and Bitcoin Layer-2 support.
- Hardened key management with AES-GCM encryption under PBKDF2-SHA256 key derivation, backward-compatible decryption, local address re-derivation, and vault ownership checks before any signing session opens.
- Built a secure dApp provider API with site allow-listing, nonce replay protection, fee caps, request expiry, and a popup-only approval queue to defend against phishing and accidental signing.
- Built a testnet faucet service with HD-wallet UTXO management, OAuth sign-in, and per-account rate limits; shipped **1,900+ automated tests** (unit, integration, and Playwright E2E) across 32K+ lines of TypeScript.

**AI Memory Protocol (Intelligence Continuity Layer)**

- *Tech Stack: Python, LangChain, ChromaDB, Redis, SQLite, TypeScript*
- Built a persistent, local-first knowledge layer to eliminate context fragmentation across siloed AI platforms (ChatGPT, Claude, Gemini), ensuring user preferences and historical data persist and remain searchable across every session.
- Implemented episodic and semantic memory layers using LangMem SDK and LangGraph Store for conversation persistence, fact extraction, and vector-backed retrieval.
- Achieved **instant semantic retrieval under 200ms** across databases exceeding 10,000+ entries using ChromaDB with OpenAI embeddings; delivered **100% data retention** across siloed AI sessions.
- Delivered a **40% reduction** in manual context re-entry for power users; designed local-first architecture with ChromaDB/SQLite and production PostgreSQL migration path.
- Implemented pay-per-use memory storage monetization using x402 protocol and ERC-8004 standard.

**AI-Powered DevOps Console & CLI (Infrastructure Automation)**

- *Tech Stack: Go (Golang), Docker, GitHub Actions / GitLab CI, Snyk / Trivy, Prometheus / Grafana, GKE / Terraform*
- Developed an autonomous "code-to-production" deployment engine with integrated security scanning and self-healing infrastructure triggers for a California-based high-growth technology firm in the Silicon Valley ecosystem.
- Engineered CLI tooling in Go for end-to-end deployment orchestration across GKE clusters with Terraform-managed infrastructure and automated CI/CD pipelines.
- Achieved **90% reduction** in deployment-related downtime and **5x faster** shipping cycles, moving client from bi-weekly to daily production deployments.
- Implemented **100% SOC2-aligned compliance** for all US-bound infrastructure by baking Snyk/Trivy security scanning into every deployment gate.

**L2GPT: Bitcoin L2 RAG Chatbot**

- *Tech Stack: Python, LangChain, FAISS, Claude Sonnet 3.5, Streamlit*
- Developed a RAG-based chatbot providing accurate, context-aware answers across all Bitcoin Layer 2 solutions and scaling technologies using LangChain and FAISS vector retrieval.
- Integrated Claude Sonnet 3.5 for high-quality technical language generation; built interactive Streamlit interface with real-time query processing.

**Stealth Market-Maker Agent (DeFAI)**

- *Tech Stack: Next.js 15, TypeScript, Python, FastAPI, LangGraph, Anchor (Rust), ShadowWire, Helius RPC, OpenAI GPT-4*
- Architected an AI-orchestrated batching layer for prediction markets eliminating front-running and alpha leakage via off-chain aggregation and ZK-shielded settlement.
- Integrated LangGraph for multi-step agent orchestration and GPT-4 for market intelligence; built smart contracts in Anchor (Rust) on Solana with MagicBlock ER for high-throughput execution.

**AI Travel Planner**

- *Tech Stack: Python, LangChain, FAISS, ELK Stack, Groq API, Streamlit, Docker, GCP, Kubernetes*
- Built an intelligent travel planning system with LangChain RAG pipelines, FAISS semantic search, and ELK Stack for comprehensive logging and analytics.
- Deployed containerized application on GCP with Kubernetes orchestration and automated CI/CD pipeline for continuous delivery.

**IoT Air Quality & Decentralized Climate Infrastructure**

- *Tech Stack: ESP32 (Embedded C), IPFS, React, Leaflet, x402 Protocol, ERC-8004*
- Designed a decentralized climate tech stack monitoring air quality via IoT devices, processing **2.6M data points per month** with a native incentivization layer for data contributors.
- Engineered immutable IPFS data layer for tamper-proof storage and implemented pay-as-you-go access monetization via x402 protocol and ERC-8004.

## Education

**PES College of Engineering** **2021 – 2025**

*Bachelor of Engineering in Computer Science* *Mandya, India*